

SIMATS ENGINEERING

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	BTech CSE BioSciences	Date:	22.08.2026

ANNEXURE A COMPARATIVE ANALYSIS

- 1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
- 2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
- 3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
- 4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
- 5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

*I S Susmitha (192571068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Name	Register Number	Course Code	Course
S Susmitha	192571068	CSA1202	Computer Architecture

Q1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Overview

Modern processors use a multi-level cache hierarchy (L1, L2, L3) to bridge the large speed gap between the CPU core and main memory (DRAM). Each level trades capacity against speed: caches closer to the core are smaller and faster, while caches farther away are larger but slower.

Comparative Analysis

Parameter	L1 Cache	L2 Cache	L3 Cache
Location	Inside each core (split I-cache/D-cache)	Per-core or shared by a small core cluster	Shared across all cores on the chip
Typical Size	32 KB – 64 KB	256 KB – 1 MB	8 MB – 32 MB
Latency	~1–4 CPU cycles (~1 ns)	~10–12 cycles (~3–5 ns)	~30–50 cycles (~10–20 ns)
Bandwidth	Highest (100+ GB/s per core)	High, lower than L1	Moderate, shared among cores
Throughput	Very high per-core throughput	High, slightly reduced due to larger array	Lower than L1/L2 due to sharing and distance
Construction	SRAM, fastest and most expensive per bit	SRAM, denser than L1	SRAM, largest and slowest of the three

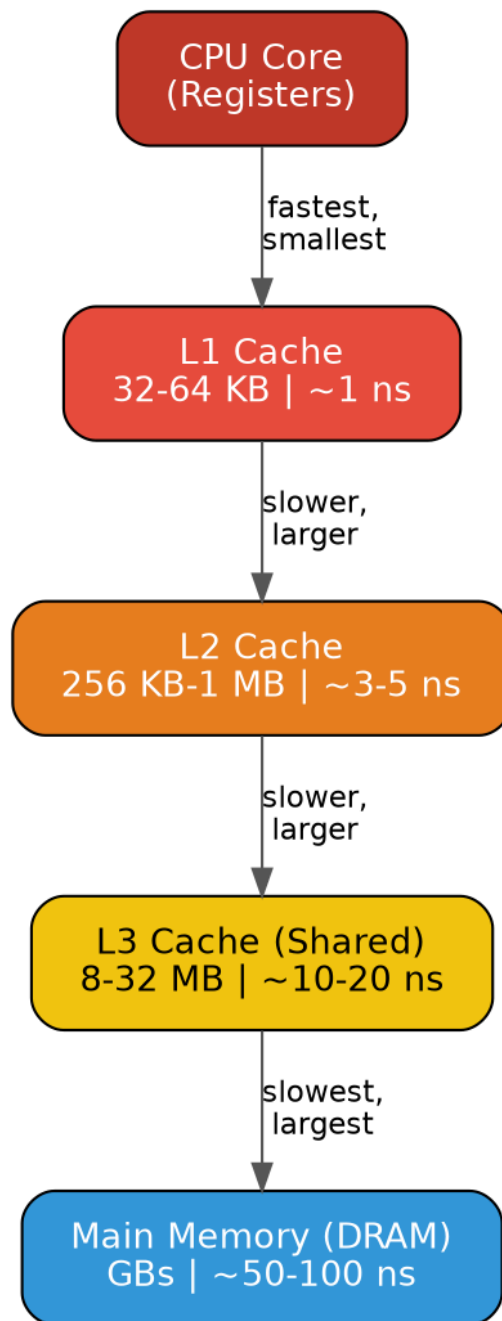


Fig 1: Cache Hierarchy — speed decreases and size increases from L1 to Main Memory

Impact on Processor Performance

L1 cache directly determines instruction and data fetch speed for the pipeline; a high L1 hit rate keeps the core running close to its peak issue rate. L2 acts as a buffer that absorbs L1 misses without forcing an expensive trip to main memory, so its size and latency strongly influence per-core performance in memory-intensive workloads. L3, being shared, plays a critical role in multi-core systems by enabling inter-core data sharing and reducing redundant main-memory accesses; however, its higher latency means that an L3 miss (going to DRAM) is the most expensive event in the hierarchy, often costing 200+ cycles. Overall, the hierarchy works because most programs exhibit temporal and spatial locality,

allowing a small, fast L1 to satisfy the majority of accesses while L2 and L3 progressively catch the remainder at increasing but still-tolerable latency.

Q2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Overview

SRAM (Static RAM) and DRAM (Dynamic RAM) are the two dominant semiconductor memory technologies. SRAM is used in caches for its speed, while DRAM is used as main memory for its density and cost efficiency.

Comparative Analysis

Parameter	SRAM	DRAM
Cell Structure	6-transistor flip-flop cell	1 transistor + 1 capacitor cell
Latency	Very low (~1–2 ns)	Higher (~50–100 ns)
Bandwidth	Very high (wide on-chip buses)	Moderate, limited by off-chip bus and refresh overhead
Throughput	High sustained throughput, no refresh stalls	Lower sustained throughput due to periodic refresh cycles
Refresh Requirement	None (bistable, holds state as long as powered)	Required every ~64 ms to prevent charge leakage
Density / Cost per bit	Low density, expensive per bit	High density, inexpensive per bit
Typical Use	L1/L2/L3 caches, registers	Main memory (DIMMs)

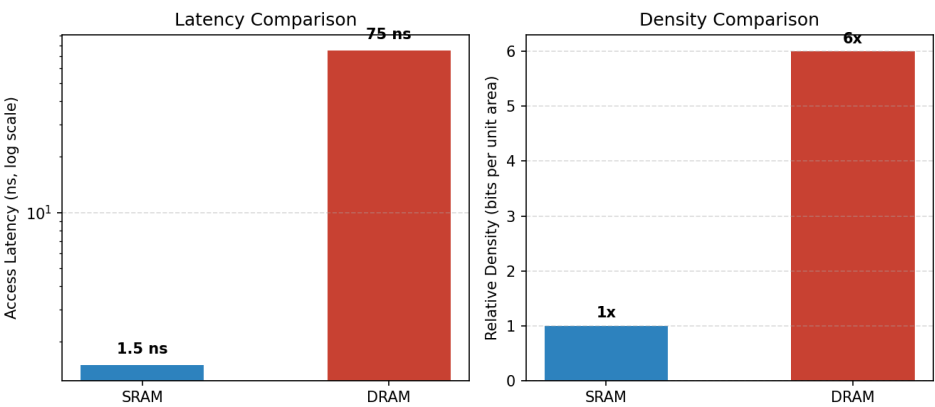


Fig 2: SRAM vs DRAM — access latency (log scale) and relative density

Evaluation

In cache design, SRAM's sub-nanosecond access time and absence of refresh stalls make it the only viable choice for keeping pace with core clock speeds. DRAM's capacitor-based cell allows far greater density, so system designers use it for main memory where capacity (gigabytes) matters more than raw speed. The trade-off directly shapes the memory hierarchy: SRAM caches hide DRAM's higher latency by serving the majority of accesses locally, while DRAM's cost-per-bit advantage makes large main memories economically feasible.

Q3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Overview

Both virtual memory (with paging) and cache memory are hierarchical techniques that give the illusion of a larger, faster memory space than what is physically fast to access. They operate at different levels of the hierarchy and are managed differently — cache primarily in hardware, virtual memory jointly by hardware (MMU/TLB) and the operating system.

Comparative Analysis

Parameter	Cache Memory	Virtual Memory (Paging)
Purpose	Bridges CPU–main memory speed gap	Bridges main memory–secondary storage capacity gap
Managed By	Hardware (cache controller)	Operating system + MMU/TLB (hardware-assisted)
Unit of Transfer	Cache line (typically 64 bytes)	Page (typically 4 KB)
Access Latency on Hit	~1–20 ns depending on level	~100 ns (main memory access via TLB hit)
Access Latency on Miss/Fault	Tens to hundreds of cycles (fetch from next level)	Milliseconds (disk/SSD page fault) — orders of magnitude slower
Throughput Impact	Miss penalty is small relative to CPU cycle time	A page fault can stall a process for a very long time, but faults are rare with good locality
Replacement Policy	LRU, LFU, random (implemented in hardware)	LRU-approximation (e.g., clock algorithm), implemented in software

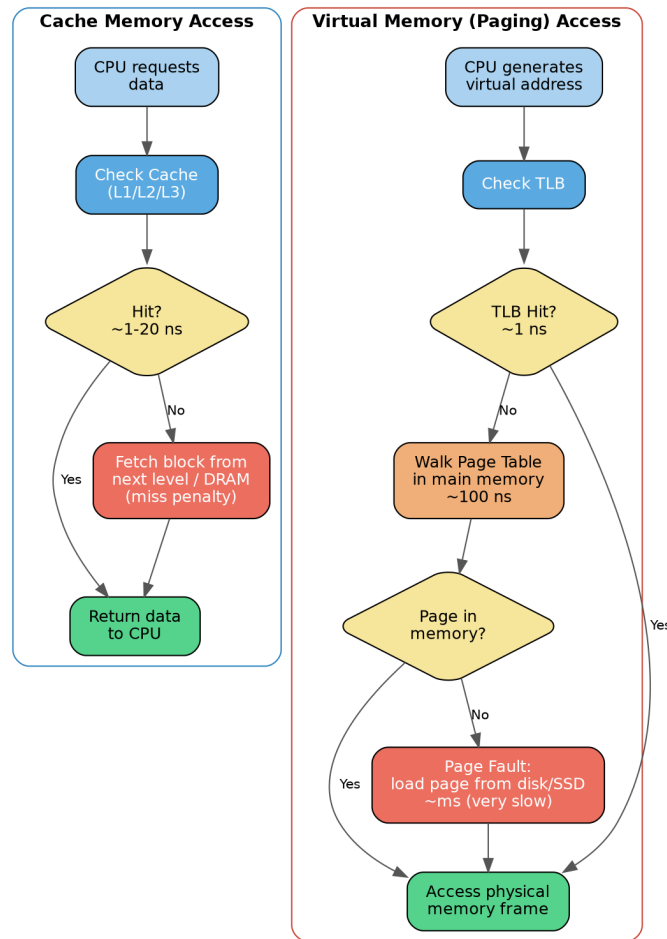


Fig 3: Cache access flow vs page-table walk / page-fault flow in paging

Critical Comparison

The two mechanisms are conceptually similar — both exploit locality of reference and rely on a fast/small store backed by a slow/large store — but they operate on vastly different timescales. A cache miss costs nanoseconds and is largely invisible to software, whereas a page fault costs milliseconds because it involves mechanical or flash storage access and OS intervention, making it roughly 10^5 – 10^6 times more expensive than a cache miss. Consequently, system throughput is far more sensitive to the page-fault rate than to the cache-miss rate: even a small increase in paging activity (thrashing) can collapse throughput, while cache misses are smoothed out by the multi-level cache hierarchy discussed in Q1.

Q4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Overview

NAND Flash and DRAM occupy different tiers of the storage/memory hierarchy. DRAM is volatile working memory, while NAND Flash is non-volatile storage used in SSDs, increasingly also explored as a persistent tier below DRAM.

Comparative Analysis

Parameter	NAND Flash	DRAM
Volatility	Non-volatile (retains data without power)	Volatile (loses data without power)
Latency	~10–100 μ s (read), higher for program/erase	~50–100 ns — roughly 100–1000 $\times$ faster than NAND
Bandwidth	Hundreds of MB/s to a few GB/s per channel (NVMe SSDs)	Tens of GB/s per channel
Endurance	Limited program/erase cycles (wear leveling required)	Effectively unlimited write endurance
Density / Cost per bit	Very high density, low cost per bit	Lower density than NAND, higher cost per bit
Access Granularity	Page/block based (must erase before rewrite)	Byte-addressable
Role in Hierarchy	Secondary storage / persistent tier	Main (working) memory

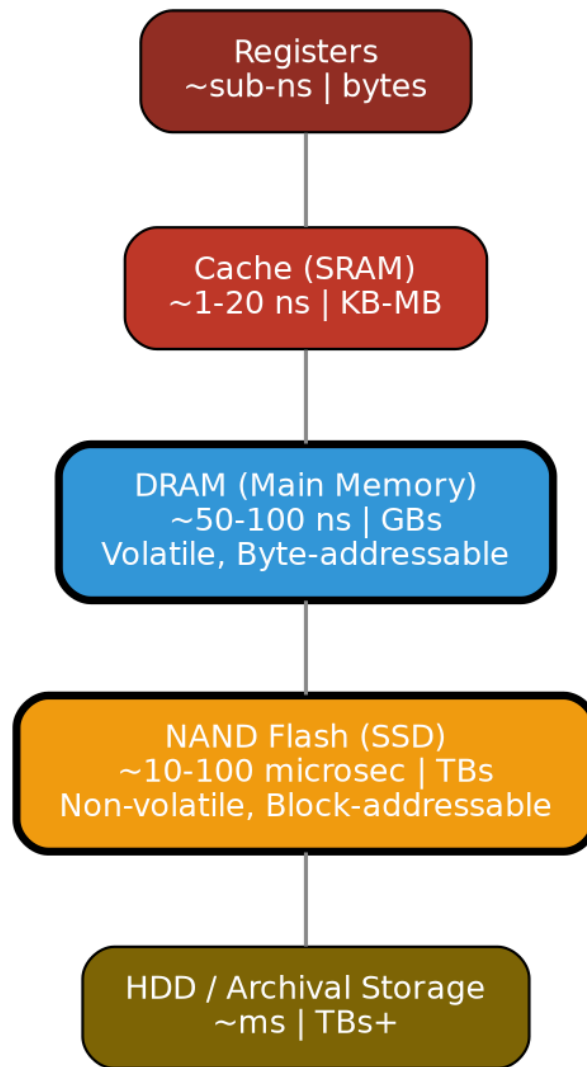


Fig 4: Memory-storage hierarchy showing DRAM and NAND Flash positioning

Suitability for Hierarchical Memory Systems

DRAM's byte-addressability and nanosecond latency make it well suited to sit directly below the cache hierarchy as main memory, actively holding the working set of running programs. NAND Flash's much higher latency and block-erase requirement make it unsuitable as a direct DRAM replacement, but its non-volatility and low cost per bit make it ideal as a persistent storage tier — and, in emerging designs, as a large 'memory-extension' layer (e.g., storage-class memory) that trades some latency for far greater capacity than DRAM can economically provide.

Q5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Overview

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM) are emerging non-volatile memory (NVM) technologies positioned to close the gap between fast-but-volatile DRAM/SRAM and dense-but-slow NAND Flash.

Comparative Analysis

Parameter	MRAM	RRAM
Storage Mechanism	Magnetic tunnel junction — resistance set by magnetic orientation	Resistance switching via formation/rupture of a conductive filament in a metal-oxide layer
Access Time	~2–20 ns (STT-MRAM), close to SRAM/DRAM speed	~10–50 ns, slightly slower and less uniform than MRAM
Throughput	High and consistent, symmetric read/write speed	Moderate; write speed can be lower/less consistent than read
Endurance	Very high ($10^{12}+$ cycles), near-DRAM-class	Lower than MRAM (10^6 – 10^9 cycles typically)
Energy Efficiency	Low read energy; write energy moderate (current-driven switching)	Low read/write energy per bit, but filament variability can raise effective energy use
Non-volatility	Yes	Yes
Maturity / Use Case	Embedded/cache-replacement, industrial and automotive NVM	Emerging, promising for storage-class memory and neuromorphic/in-memory computing

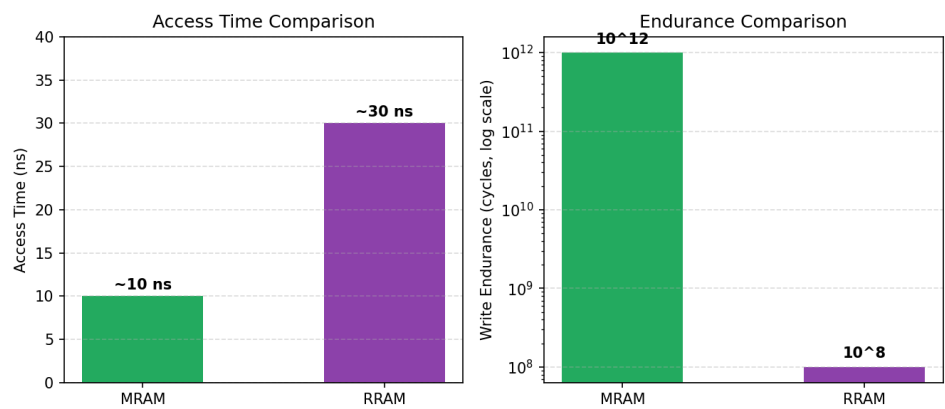


Fig 5: MRAM vs RRAM — access time and write endurance comparison

Performance Analysis

MRAM currently offers the more mature and predictable performance profile: near-SRAM access times, symmetric read/write behavior, and very high endurance make it attractive for replacing SRAM or DRAM in energy-sensitive embedded and cache applications. RRAM's simpler cell structure gives it a potential density and cost advantage, and its resistance-based switching is attractive for in-memory computing, but variability in filament formation currently limits its endurance and write consistency compared to MRAM. In terms of energy efficiency, both technologies benefit from zero standby leakage

(unlike SRAM/DRAM, which consume static/refresh power), but MRAM's more deterministic switching generally gives it an edge in write-energy predictability, while RRAM's lower cell complexity gives it an edge in per-bit fabrication energy and cost at scale.